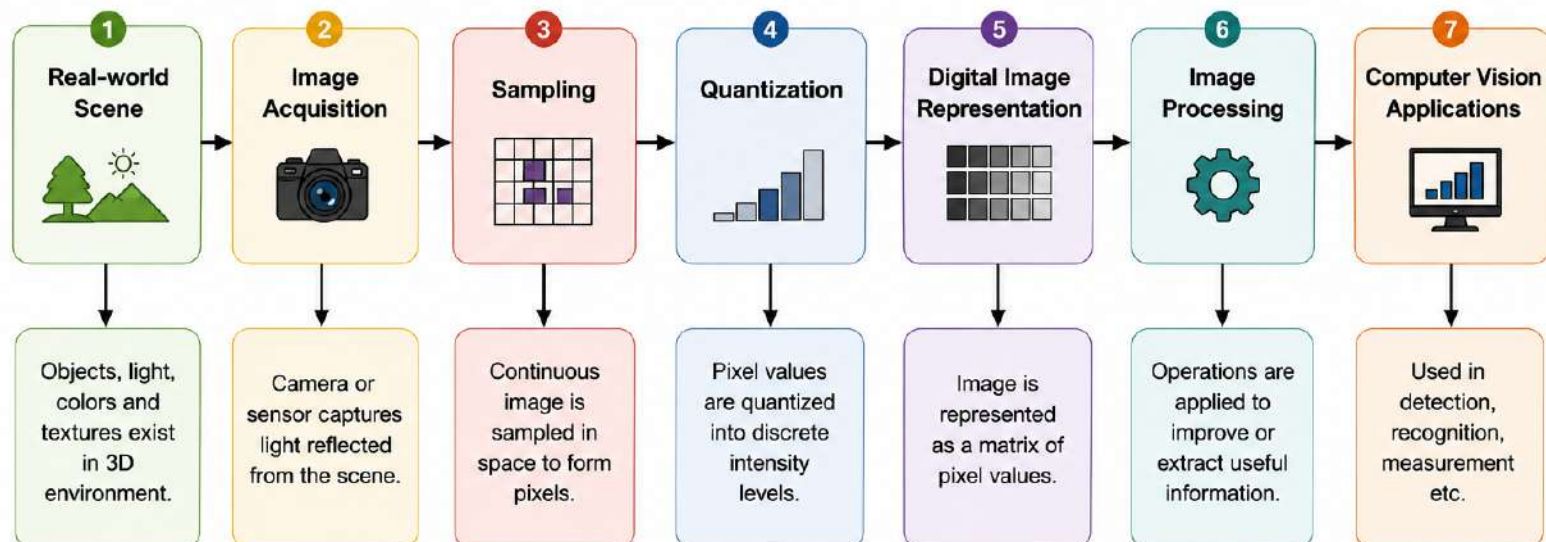


1st Question

Concept Map – From Real-world Scene Capture to Digital Image Representation

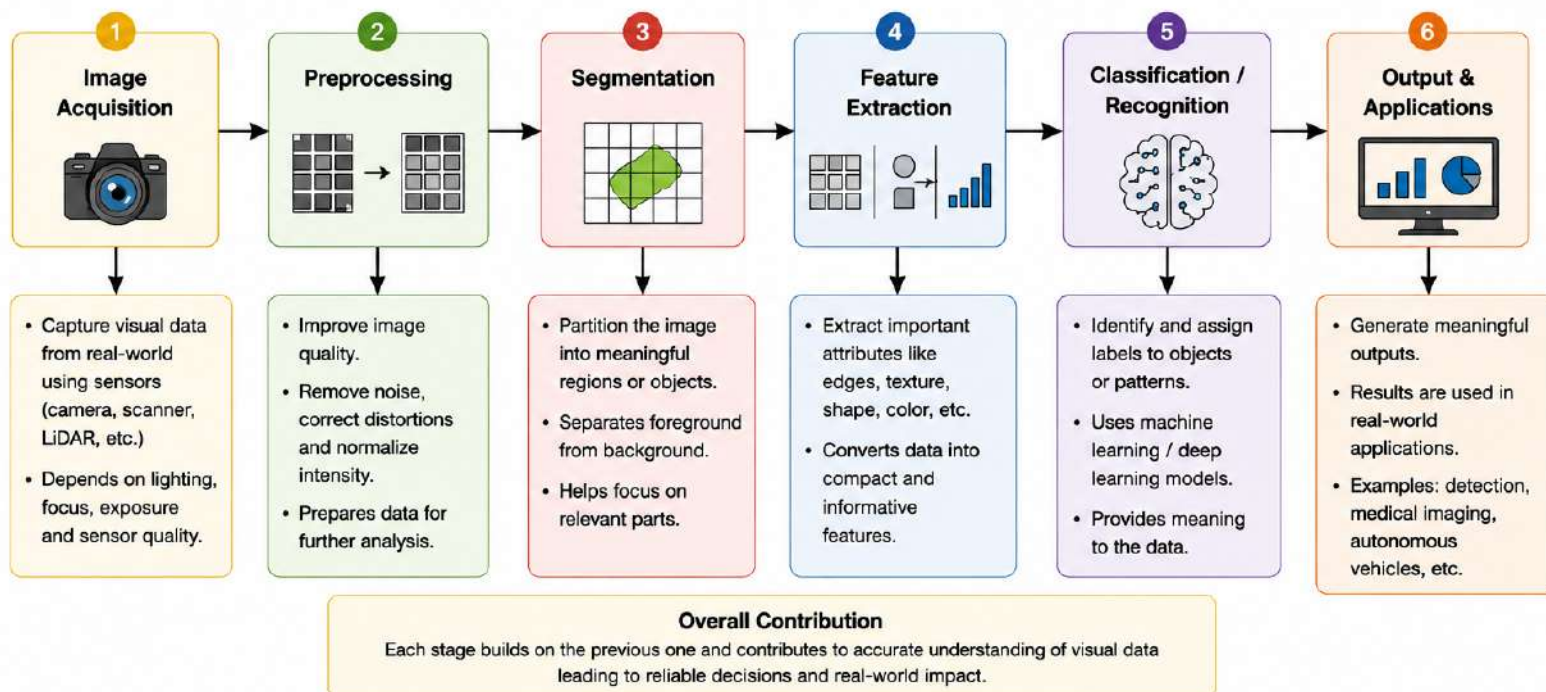


References

- Gonzalez & Woods, Digital Image Processing, 4th Edition.
- Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition.
- OpenCV Documentation, <https://docs.opencv.org>

2nd Question

Concept Map – Stages of Visual Data Handling in a Computer Vision System



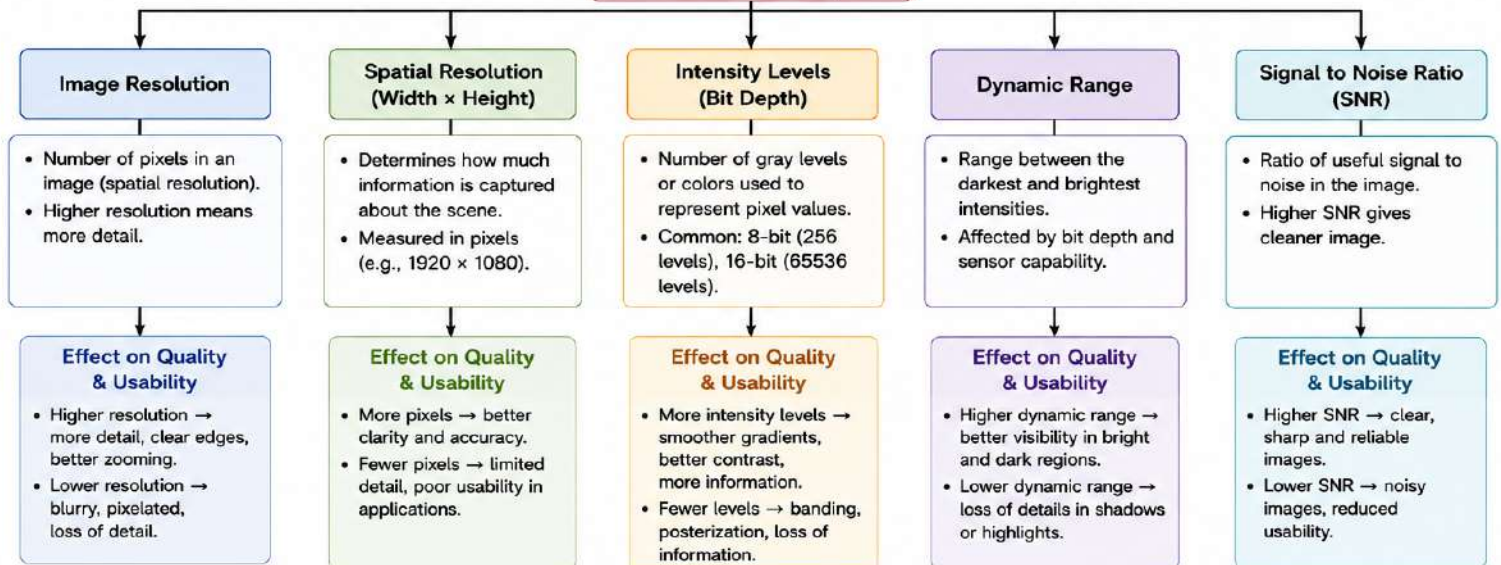
References:

- Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd Edition, Springer.
- Rafael C. Gonzalez & Richard E. Woods, *Digital Image Processing*, 4th Edition, Pearson.
- OpenCV Documentation, <https://docs.opencv.org/>

3rd Question

Concept Map – Image Resolution and Intensity Levels

Image Quality and Usability



Overall: Higher resolution and more intensity levels with good dynamic range and high SNR lead to better image quality, accurate analysis and wider usability in computer vision applications.

References

1. Rafael C. Gonzalez & Richard E. Woods, *Digital Image Processing*, 4th Edition, Pearson.
2. Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd Edition, Springer.
3. Jain, A. K., *Fundamentals of Digital Image Processing*, Prentice Hall.

4th Question

Concept Map – Image Enhancement and Restoration Techniques

Image Enhancement (Improves Visual Quality)

1. Contrast Enhancement

- Improves difference between darker and brighter regions.
- Makes image more clear and easy to interpret.

2. Histogram Equalization

- Spreads intensity values over full range.
- Improves global contrast.

3. Sharpening

- Enhances edges and fine details.
- Makes image appear sharper.

Image Restoration (Removes Degradations)

4. Noise Removal (Filtering)

- Removes unwanted noise from image.
- Examples: Mean filter, Median filter, Gaussian filter.

5. Deblurring (Restoration)

- Removes blur caused by camera motion or out-of-focus.
- Restores clear details.

6. Inpainting

- Removes missing or damaged parts.
- Fills the area using surrounding information.

Summary

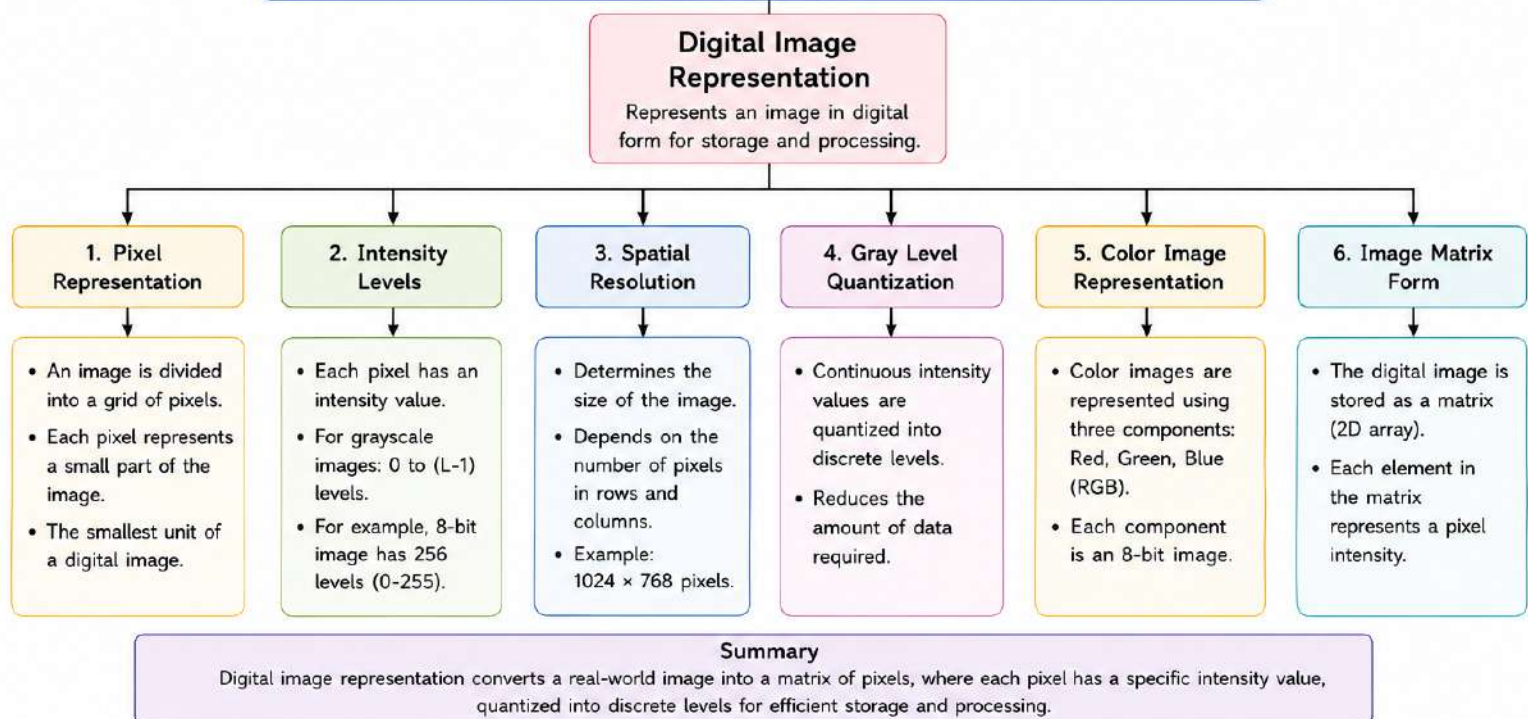
Enhancement techniques improve the visual quality for better viewing, while restoration techniques remove degradations to recover the original image for accurate analysis.

References:

1. Rafael C. Gonzalez & Richard E. Woods, *Digital Image Processing*, 4th Edition.
2. Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd Edition.
3. Jain, A. K., *Fundamentals of Digital Image Processing*, Prentice Hall.

5th Question

Concept Map – Digital Image Representation



References:

1. Rafael C. Gonzalez & Richard E. Woods, *Digital Image Processing*, 4th Edition, Pearson.
2. Jain, A. K., *Fundamentals of Digital Image Processing*, Prentice Hall.
3. Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd Edition.